

GRADE X - MATHEMATICS WORKSHEET (2026-27)

Extracted questions from the two uploaded worksheet photographs. Handwritten workings/answers have been omitted.

SECTION A - Multiple Type Questions

1. If the equation $(k - 1)x^2 + kx + 1 = 0$ has equal and real roots, what is the value of k ?
 - A. 0
 - B. 4 only
 - C. 0 or 4
 - D. 2
2. If α and β are roots of $2x^2 - 5x + 3 = 0$, what is the value of $\alpha^2 + \beta^2$?
 - A. $19/4$
 - B. $13/4$
 - C. $7/4$
 - D. $25/4$
3. If α and β are the roots of the equation $x^2 - p(x + 1) - c = 0$, then $(\alpha + 1)(\beta + 1) =$
 - A. c
 - B. $c - 1$
 - C. $1 - c$
 - D. $1 + c$
4. If α and β are roots of $x^2 - 4x + 1 = 0$, find the value of $\alpha^4 + \beta^4$.
 - A. 194
 - B. 196
 - C. 198
 - D. 192
5. Which of the following is a solution of the equation $x^2 - 6x + 5 = 0$?
 - A. 2
 - B. 5
 - C. 9
 - D. 15
6. Which of the following is a quadratic equation?
 - A. $x^3 + 2x + 1 = 0$
 - B. $x^2 + 1/x = 3$
 - C. $(x + 1)(x - 2) = x^2 - 3$
 - D. $2x^2 - 5x + 3 = 0$
7. The discriminant of the quadratic equation $(x + 5)^2 = 2(5x - 3)$ is:
 - A. 5
 - B. 124
 - C. -5
 - D. -124

8. The quadratic equation $2x^2 - \sqrt{5}x + 1 = 0$ has
- A. two distinct real roots
 - B. two equal real roots
 - C. two equal real roots
 - D. two non-real roots
9. If the equation $x^2 - mx + 1 = 0$ does not possess real roots, then
- A. $m > 2$
 - B. $m < -2$
 - C. $-2 < m < 2$
 - D. $-3 < m < 3$
10. Which of the following has 2 as a root?
- A. $x^2 - 4x + 5 = 0$
 - B. $x^2 - 3x - 12 = 0$
 - C. $2x^2 - 7x + 6 = 0$
 - D. $3x^2 - 6x - 2 = 0$
11. A lad was asked his age by his friend. The lad said, "The number you get when you subtract 25 times my age from twice the square of my age will be thrice your age." If the friend's age is 14, then the age of the lad (in years) is:
- A. 21
 - B. 28
 - C. 14
 - D. 25
12. The roots of the equation $x^2 - 5x + 6 = 0$ are
- A. 2 and 3
 - B. -2 and -3
 - C. 1 and 6
 - D. -1 and -6
13. If the sum of the roots is 6 and product is 8, the quadratic equation is:
- A. $x^2 + 6x + 8 = 0$
 - B. $x^2 - 6x + 8 = 0$
 - C. $x^2 - 8x + 6 = 0$
 - D. $x^2 + 8x - 6 = 0$
14. Assertion: The equation $2x^2 + 3x + 1 = 0$ has two distinct real roots.
Reason: A quadratic equation $ax^2 + bx + c = 0$ has two distinct real roots when its discriminant $D = b^2 - 4ac > 0$.
15. Assertion: Every quadratic equation has exactly two roots.
Reason: A quadratic polynomial of degree 2 can have at most two zeros.

SECTION B - Short Answer Type Questions

16. For what value of k does $kx^2 - 6x + 1 = 0$ have equal roots?
17. If one root of $x^2 - 5x + k = 0$ is 2, find the value of k .

18. A train travels 360 km at a uniform speed. If its speed had been 5 km/h more, it would have taken 1 hour less. Form a quadratic equation for this.
19. For what values of k does $(k - 1)x^2 + kx + 1 = 0$ have equal roots?
20. Solve: $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ by factorization.
21. A ball is thrown upward from the ground. Its height (in metres) after t seconds is given by $h = -5t^2 + 20t$. Find the time(s) at which the ball is at a height of 15 metres. Also find the maximum height reached by the ball.

SECTION B - Short Answer Type Questions (continued)

22. A shopkeeper buys a number of books for ₹1200. If he had bought 10 more books for the same amount, each book would have cost ₹20 less. Form and solve a quadratic equation to find the original number of books bought.

23. The sum of ages of a father and his son is 45 years. Five years ago, the product of their ages was 124. Form a quadratic equation and find their present ages.

24. A two-digit number is such that the product of its digits is 14. When 45 is added to the number, the digits interchange. Find the two-digit number.

SECTION C - Long Answer Type Questions

25. Two water pipes together fill a tank in $9\frac{3}{8}$ minutes. The larger pipe alone fills it 10 minutes faster than the smaller pipe. Find the time each pipe takes to fill the tank individually. If the smaller pipe starts 5 minutes earlier and then both pipes work together, how long will the combined phase last?

26. In a class test, the sum of Shefali's marks in Mathematics and English is 30. Had she got 2 marks more in Mathematics and 3 marks less in English, the product of their marks would have been 210. Find her marks in the two subjects.

27. A train covers a distance of 480 km at a uniform speed. If the speed had been 8 km/h less then it would have taken 3 hours more to cover the same distance. Find the usual speed of the train.

28. A peacock is sitting on the top of a pillar, which is 9 m high. From a point 27 m away from the bottom of the pillar, a snake is coming to its hole at the base of the pillar. Seeing the snake, the peacock pounces on it. If their speeds are equal, at what distance from the hole is the snake caught?

29. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/h and the time of flight increased by 30 minutes. Find the duration of flight.

30. One pipe can fill a tank in $(x + 2)$ hours and the other pipe can fill the same tank in $(x + 7)$ hours separately. If both the pipes, when opened together, take 6 hours to fill the empty tank, find the value of x .

SECTION D - Case Based Questions

31. A ball is thrown upward from a rooftop. Its height (in metres) above the ground after t seconds is given by $h(t) = -5t^2 + 20t + 25$. Riya wants to find when the ball hits the ground and when it reaches maximum height.

- What is the height of the rooftop from the ground?
- Set up the equation to find when the ball hits the ground. Simplify to standard form.
- Find the time at which the ball is at a height of 40 m. How many solutions exist? What does this mean physically?

32. A farmer has 80 m of fencing wire to enclose a rectangular field along a straight river. Since the river forms one side of the field, only three sides need fencing. The farmer wants to maximise the area of the field. His son Ajay sets up a quadratic equation to model the situation.

- If the width of the field (perpendicular to the river) is x m, express the length in terms of x . Write the area A as a quadratic function of x .
- For what value of x is the area equal to 750 m^2 ? How many solutions exist and what do they represent physically?

iii) What is the maximum area the farmer can enclose? Ajay claims it is 800 m^2 . Verify using the vertex of the parabola. Is a quadratic equation helpful here?